

Hydrogen-Fired Power Generation — OEM Survey

MYSGH2 Project (Malaysia → Singapore Green Hydrogen via Green Ammonia) | Gas turbines, 2-stroke & 4-stroke gas engines | Last verified 2026-09-10

OEM	Gas Turbine (hydrogen-fired solution/status)	2-Stroke Gas Engine (hydrogen-fired solution/status)	4-Stroke Gas Engine (hydrogen-fired solution/status)	News link
GE Vernova	Commercial blends + validated 100 % H2 combustor. HA-class (7HA/9HA): OEM states up to 50 % H2 today with DLN 2.6e dry-low-NOx combustion, "technology pathway to 100 %". B-/E-class: full-scale 100 % H2 DLN combustor validation campaign completed 15 Jan 2025 at Greenville, SC (full-size 6B DLN prototype at pressure/flow/temperature), NOx <25 ppm dry, no water/diluent , +4-7 % CC heat-rate benefit; DLN system planned available for new & existing B/E units as early as 2026 . Aeroderivative: LM6000VELOX package announced as 100 % H2-capable , first customer ATCO/BOC Whyalla, South Australia (4 units, 200 MW peaker) , announced at COP29 19 Nov 2024 , commissioning slated early 2026 .	No known programme (GE Vernova does not make reciprocating engines).	No known programme (Jenbacher/Waukesha are not GE Vernova assets -- see INNIO).	https://www.gevernova.com/news/press-releases/ge-vernova-validates-100-hydrogen-fueled-dln-combustor-technology-aiming-6 https://www.powermag.com/ge-vernova-unveils-100-hydrogen-fueled-aeroderivative-gas-turbine-solution-secures-first-customer/
Siemens Energy	Broadest tested portfolio; 100 % demonstrated on industrial frame. SGT-400 (~13 MW industrial) ran on up to 100 % green H2 in the EU HYFLEXPOWER power-to-H2-to-power demo at Saillat-sur-Vienne, France (announced 12 Oct 2023) -- 1 MW electrolyser, 1 t H2 store; 0-100 % H2 blends demonstrated. SGT-800 rated up to 75 vol % H2 ; first plant concept to receive TÜV SÜD "H2-Ready" certification. HL-class (SGT5-/SGT6-9000HL) rated H2-ready to 50 vol % , 25 premix burners; "Mission H2 Power" with SSE (launched 9 Dec) targets a 100 % H2 combustion system for SGT5-9000HL at Keadby 2 (849.45 MW). Company roadmap: 100 % H2 capability for new turbines by 2030 .	No known programme.	No known programme.	https://www.siemens-energy.com/global/en/home/press-releases/hyflexpower-consortium-successfully-operates-a-gas-turbine-wit-h-.html https://www.powermag.com/siemens-energy-joins-forces-for-bold-100-hydrogen-hl-gas-turbine-leap/
Mitsubishi Power / MHI	Large-frame blends demonstrated grid-connected; 100 % on mid-size frame. M501JAC (1,650 degC class): 30 vol % H2 co-firing demonstrated at part and full load, grid-connected , at T-Point 2, Takasago (Nov 2023), with a DLN combustor ; later raised to 50 vol % (~22 % CO2 cut at 50 % H2). OEM roadmap: advanced-class fleet 30 % today -> 50 % from mid-2025 -> 100 % by 2030 . H-25 (30-40 MW class): 100 % hydrogen firing test performed 2024 at Takasago Hydrogen Park (unit installed as a compressor driver); OEM development philosophy stated as validation of 100 % H2 in 2024, commercialisation ~2025 -- *no post-2024 confirming update located; treat commercialisation date as unconfirmed.*	No known programme.	No known programme located for a hydrogen-fired MHI reciprocating gas engine -- *not the same as confirmed absence; flagged for verification.*	https://www.mhi.com/news/23113001.html https://gasturbineworld.com/mitsubishi-power-hydrogen-gas-turbine-technology/
Ansaldo Energia	70 % released commercially; 100 % demonstrated on rig. GT36 (H-class, sequential/constant-pressure sequential combustion): 70 % H2 in NG blends released for operation , development to 100 % under way; under EU FLEX4H2 , a hydrogen-optimised GT36 combustor demonstrated 0->100 % H2 at high pressure . GT36 has received TÜV SÜD Hydrogen Readiness certification (reported 1 Sep 2026). AE94.3A : designed for co-firing up to 40 % H2 with instant revert to 100 % CH4; two units on a 25 % H2 mix in commercial operation since 2006 , >300,000 EOH. GT26 : hydrogen blending across a wide fuel range without hardware change.	No known programme.	No known programme.	https://www.ansaldoenergia.com/about-us/media-center/power-generation-news-insights/detail-news/gt36-sequential-combustion-technology-achieves-100-hydrogen https://fuelcellworks.com/2026/09/01/h2/ansaldo-energia-gt36-gains-hydrogen-readiness-certification-as-ae94-3a-passes-hvo-validation
Baker Hughes	100 % H2 on a commercial industrial frame, no pilot fuel. NovaLT16 (~16-17.5 MW class): OEM states it can start up and burn gas blends up to 100 % hydrogen , switch between NG, blends and 100 % H2 with no hardware changes and without pilot fuel . Dedicated hydrogen test facility, Florence, Italy with full-load test bench to 100 % H2 ; one NovaLT16 full-load tested and delivered to Air Products' Net-Zero Hydrogen Energy Complex, Edmonton, Canada . (Aeroderivative PGT25+/LM2500+ line is a separate, NG-basis product family -- no 100 % H2 claim located.)	No known programme.	No known programme.	https://www.bakerhughes.com/gas-turbines/novalt-technology/novalt16 https://gasturbineworld.com/baker-hughes-hydrogen-turbines/
Rolls-Royce (Power Systems / mtu)	No known stationary power-gen H2 gas turbine programme. (Rolls-Royce's 100 %-H2 turbine work is aero -- AE2100/Pearl with easyJet -- *not* power generation; flagged so it is not mis-scoped.)	No known programme.	Commercial H2-ready; 100 % H2 installations. mtu Series 4000 FNER/FV gas engines hold TÜV SÜD H2-readiness certification (announced 11 Jul 2024) covering newbuild and retrofit power plants at up to 25 vol % H2 blend and 100 vol % operation . First 100 % hydrogen mtu 4000 FNER/FV installation planned early 2025 at two CHP plants for the Enerport II project (Duisburg). New 20-cyl mtu Series 4000 L64 (60 Hz, 2.8 MW , full output in 45 s) from 2026 , stated operable on 100 % H2 or blends.	https://www.rolls-royce.com/media/press-releases/2024/11-7-2024-energy-transition-rr-reaches-milestone-with-h2-ready-certificate-for-mtu-engines.aspx https://www.rolls-royce.com/media/press-releases/2025/02-10-2025-rr-introduces-fast-start-mtu-gas-gensets-for-powering-data-centers.aspx
Solar Turbines (Caterpillar)	Long high-H2 field record; no 100 % product claim located. Solar states experience with hydrogen-rich fuels since 1985 , >2 million operating hours, with some units on as much as 80 % hydrogen content ; Taurus 60 and Titan 130 fleet of 46 units has exceeded 2 million operating hours on high-hydrogen fuel . 2024 : operated a gas turbine on a 60 % hydrogen blend with Chevron. Retrofit/upgrade path offered for hydrogen-blend operation. *No source located stating a commercially released 100 % H2 Solar product -- flagged as a gap, not as absence.*	No known programme.	n/a for this OEM (recip engines sit under Caterpillar -- see next row).	https://www.solarturbines.com/en_US/about-us/news-and-press-releases/converting-high-hydrogen-fuel-to-electricity.html https://www.turbomachinerymag.com/view/vendor-spotlight-solar-turbines

OEM	Gas Turbine (hydrogen-fired solution/status)	2-Stroke Gas Engine (hydrogen-fired solution/status)	4-Stroke Gas Engine (hydrogen-fired solution/status)	News link
Caterpillar (Cat Electric Power, recip)	See Solar Turbines row.	No known programme.	Commercial blend portfolio + 100 % H2 demonstrator. Cat gas genset portfolio 400 kW-4.5 MW offers hydrogen blending via factory-installed hardware and retrofit kits on CG132B, CG170B, CG260 and G3500H platforms (CG260 added to the H2-capable lineup, announced 2024). G3516 gas gensets capable of 100 % hydrogen offered as demonstrator units since 2022, max rating 1,000 kW at 50 or 60 Hz continuous.	https://www.cat.com/en_US/news/engine-press-releases/caterpillar-advances-hydrogen-program-by-adding-cat-cg260-gas-generator-set-to-lineup.html https://www.cat.com/en_US/news/engine-press-releases/caterpillar-expands-lineup-of-hydrogen-fueled-power-solutions-with-generator-sets-and-upgrade-kits.html
Everllence (ex-MAN Energy Solutions)	Not surveyed / unconfirmed -- MAN/Everllence markets MGT-series industrial gas turbines; no hydrogen programme located in this pass. Flagged, not asserted.	World-first H2 combustion on a marine 2-stroke -- R&D. Licensee MITSUI E&S converted one of four cylinders of an MAN B&W ME-GI 50-bore two-stroke to hydrogen operation and achieved stable hydrogen combustion up to 100 % load at Tamano (announced Mar 2024); GHG reduction up to 95 %, remainder from pilot fuel. No commercial hydrogen two-stroke product released.	Research engine on 100 % H2. First single-cylinder 35/44DF H2 four-stroke research engine commissioned in Augsburg, late 2025, and already run on 100 % hydrogen. Series gas engines 35/44GTS and 51/60G accept up to 25 % hydrogen blended into the gas mix.	https://www.offshore-energy.biz/man-es-and-mitsui-es-perform-worlds-first-hydrogen-combustion-test-on-marine-2-stroke-engine/ https://www.man-es.com/energy-storage/campaigns/hydrogen-power-solutions
WinGD	No known programme.	No commercial hydrogen two-stroke in the near-term roadmap. As of 2026 WinGD's released/announced alternative-fuel two-strokes are LNG (X-DF), methanol (X-DF-M), ethanol (2026 launch, deliveries 2027) and ammonia (X-DF-A -- TAT and FAT completed Jan 2026). WinGD has publicly characterised hydrogen as having limited potential as a direct marine fuel (low volumetric energy density; role mainly as a building block for other fuels). *Earlier (c. 2021) WinGD statements referencing hydrogen-capable engines could not be reconciled with a current product date -- unconfirmed.*	No known programme (WinGD is a two-stroke house).	https://wingd.com/news-media/news/wingd-to-introduce-first-ethanol-fuelled-marine-two-stroke-engine-in-2026 https://www.powerprogress.com/news/wingd-looks-to-future-fuels-for-ocean-going-ships/8024800.article
Wärtsilä	No known programme.	No known programme.	The most advanced large 100 % H2 engine. Wärtsilä 31H2 -- pure-hydrogen 4-stroke on the Wärtsilä 31 platform (bore/stroke 310/430 mm, 8-16 cylinders, 4.6-10.4 MW per unit on the 31 platform), OEM-described as the world's largest pure hydrogen engine; fuel-flexible (100 % H2, NG, or blends). June 2026: ran on 100 % hydrogen supplying power to the Spanish national grid from the Bermeo laboratory -- stated as the world's first large-scale demonstration of a hydrogen engine on 100 % pure H2. Product line announced 18 Jun 2024: 31H2 (pure H2) and 31SG-H2 (100 %-H2-ready NG engine); available for tendering 2025, delivery 2026.	https://www.wartsila.com/media/news/11-06-2026-world-s-first-large-scale-100-hydrogen-engine-tested-at-wartsila-s-bermeo-laboratory-to-support-the-spanish-grid-3760292 https://www.wartsila.com/media/news/18-06-2024-wartsila-launches-world-s-first-large-scale-100-hydrogen-ready-engine-power-plant-3464281
Kawasaki Heavy Industries (KHI)	Commercial 100 % H2 at small scale; 60 % on 30 MW frame. GPB17MMX, 1.8 MW class -- world's first 100 %-hydrogen, *dry* (DLE) combustion gas turbine cogeneration system, launched Sep 2023; Hydrogen Technology of the Year 2024. L30A (30.9 MW, 41.2 % SC thermal efficiency, DLE, 15 ppm NOx on NG): up to 60 vol % H2 co-firing with DLE; a 100 % hydrogen diffusion-flame combustor version was commissioned in Kobe (world-first for 100 % H2 operation of an industrial GT, 2020) -- note diffusion (wet/NOx-penalised) vs. dry distinction. 30 MW-class 30 % H2 DLE combustor also offered.	Partner in the world-first H2 two-stroke -- see J-ENG row: HyENG JV (J-ENG + Kawasaki + Yanmar Power Technology, formed 2021). Kawasaki is not the engine designer of record.	World-first commercial 30 % H2 co-firing large gas engine. Sales of KG-series "Green Gas Engine" capable of 30 vol % hydrogen co-firing launched 30 Sep 2025 -- stated as a world first. Verification on an 8 MW-class genset at Kobe Works, Oct 2024 -> Sep 2025; co-firing ratio adjustable during operation; minimal modification to existing KG units (KG series >240 units as of Mar 2025).	https://global.kawasaki.com/en/corp/newsroom/news/detail/?f=20250930_4865 https://global.kawasaki.com/en/energy/equipment/gas_engines/hydrogen_blended.html
Japan Engine Corporation (J-ENG)	No known programme.	World-first large low-speed 2-stroke hydrogen engine -- demonstration. 6UEC35LSGH full-scale hydrogen-fuelled two-stroke; land-based operation announced 28 Oct 2025; all-cylinder hydrogen co-firing from Mar 2026, stable combustion across all six cylinders with hydrogen >95 % of fuel energy at full load. Programme run through the HyENG JV (J-ENG, Kawasaki, Yanmar Power Technology, est. 2021). Commercial deployment expected in the 2030s.	No known programme (J-ENG is a two-stroke house).	https://www.j-eng.co.jp/en/news/lnhnoq0000005xn-att/J-ENG_Press_20251028_EN https://interestingengineering.com/energy/world-first-hydrogen-fueled-engine-cargo-ships
HD Hyundai (HHI-EMD / HiMSEN; HD Hyundai Infracore)	No known programme.	No own hydrogen two-stroke (HHI builds licensed MAN B&W and WinGD two-strokes -- e.g. WinGD's X52DF-A ammonia engine was tested at HHI, Jan 2026).	Pure-H2 4-stroke announced; power-gen engine targeted 2026. HiMSEN pure-gas four-stroke hydrogen marine engine: component manufacture Q1 2025, R&D testing from Q2 2025, sales release anticipated Dec 2025 (trade press, *The Motorship*) -- status of actual release unconfirmed. HD Hyundai Infracore roadmap: hydrogen engines commercialised for mobility 2025 and power generation 2026, with 11-litre hydrogen engines for power generation planned for mass production in 2026.	https://www.motorship.com/2-stroke-and-4-stroke/himsen-confirms-2025-pure-hydrogen-engine-launch-plans/1488666.article https://esg.hd.com/en/news/842
Doosan Enerbility	50 %-class combustor in development; 100 % targeted 2027. DGT6-300H.S1 (270 MW) and DGT6-300H.S2 (380 MW) H-class frames (>40 % / >43 % simple-cycle efficiency). With KIMM (from May 2020), developing a dual-fuel combustor for a 300 MW hydrogen gas turbine at >=50 % H2 (stated ~21.4 % emissions reduction vs LNG at a 50 % mix). Developing a 400 MW 100 %-hydrogen turbine, targeted 2027, with the combustor to be completed by 2026. Korea's first hydrogen gas turbine demonstration is reported as targeted for 2027.	No known programme.	No known programme.	https://gasturbineworld.com/doosan-gas-turbines/ https://www.powermag.com/south-korea-eyes-first-hydrogen-gas-turbine-demonstration-in-2027/
INNIO (Jenbacher)	No known programme.	No known programme.	Commercially available on 100 % H2 at Type 4; whole line from 2025. Jenbacher Type 4 (~0.5-0.9 MW) available as H2 engines running on 100 % hydrogen, or 100 % NG, or blends. All other Jenbacher engines offered with the "Ready for H2" option -- up to 25 vol % H2 in pipeline gas, convertible later to 100 % H2; the entire product line was slated for 100 % H2 rollout from 2025 (Type 4 and Type 6 / J420, J620 shown at 0-100 % H2, 2025+ in INNIO's own roadmap material). References: Hyosung Chemical Yongyeon plant, Ulsan, South Korea -- first 100 %-H2-fuelled engine pilot power plant in APAC; 3 MW 100 %-hydrogen data-centre backup power demonstration with the Net Zero Innovation Hub, completed 23 Apr 2026.	https://www.innio.com/en/news-media/press-releases/innio-and-net-zero-innovation-hub-deliver-world-first-3-mw-demonstration-of-100-hydrogen-backup-power-for-data-centers/ https://www.innio.com/en/news-media/press-releases/innio-technology-selected-for-first-100-hydrogen-engine-power-plant-in-asia-pacific/

OEM	Gas Turbine (hydrogen-fired solution/status)	2-Stroke Gas Engine (hydrogen-fired solution/status)	4-Stroke Gas Engine (hydrogen-fired solution/status)	News link
Cummins	No known programme.	No known programme.	Hydrogen ICE developed, but on-highway/off-highway -- no power-gen genset located. B6.7H 6.7 L hydrogen ICE development completed (with Johnson Matthey, PHINIA, Zircotec); 290 hp / 1,201 N·m in medium-duty package; shown at Busworld 2025. A 15-litre hydrogen ICE is in development. *No Cummins hydrogen genset/power-generation product located in this pass* -- treat any power-gen application as unconfirmed .	https://www.route-one.net/suppliers/cummins-b6-7h-hydrogen-engine-development-is-completed/ https://www.cummins.com/news/releases/2025/11/10/cummins-x15-highway-receives-powertrain-magazines-2026-alternative-engine
Hanwha (Hanwha Impact / Hanwha Power Systems / PSM / Thomassen)	Retrofit combustor route -- world-record blends on installed frames. PSM FlameSheet(TM) retrofit combustor delivered an industry-record 60 % hydrogen at baseload with single-digit ppm NOx and CO in an 80 MW-class heavy-duty frame (Frame 7E, Hanwha Impact Daesan, Korea) -- announced Jun/Jul 2023 ; Hanwha states 59.5 % H2 co-firing in an 80 MW-class turbine as a world first, with ~22 % CO2 reduction vs LNG and NOx 6 ppm . Up to 80 % H2 demonstrated in high-pressure rig testing (DLR). FlameSheet is a retrofit for GE and Siemens frames , 33 units operating/contracted. Hanwha's own materials also claim 100 % hydrogen combustion demonstrated in an 80 MW-class gas turbine -- vendor claim; date/conditions not verified in this pass.	No known programme.	No known programme.	https://www.hanwha.com/newsroom/news/press-releases/hanwha-becomes-worlds-first-to-successfully-demonstrate-a-59-5-hydrogen-co-firing-rate-in-80-megawatt-class-gas-turbines.do https://www.psm.com/news-press-release-archive/psm-thomassen-energy-and-hanwha-power-systems-achieve-record-hydrogen-use-in-gas-turbine-power-plant
Bergen Engines (Langley Holdings)	No known programme.	No known programme.	100 % H2 firing achieved; 25 % blend across the range without modification. Dec 2024: 100 % firing of a Bergen engine on hydrogen achieved; industrialisation work continuing through 2025+. Previously demonstrated a Bergen gas engine at full load on up to 30 % H2 blended with NG at an industrial customer site in Spain. Full natural-gas engine range now stated to run on 25 % hydrogen in full operation without modification .	https://www.bergenengines.com/bergen-engines-100-hydrogen-fueled-engine-on-track/ https://www.modernpowersystems.com/analysis/bergen-demonstrates-100-hydrogen/
Yanmar (Yanmar Power Technology)	No known programme.	Partner in HyENG (with J-ENG and Kawasaki) -- see J-ENG row; not the engine designer of record.	~500 kW pilot-ignition H2 4-stroke demonstrated . Land-based demonstration of a pilot-ignition hydrogen 4-stroke high-speed engine for coastal-vessel power generation completed at rated output ~500 kW (announced 30 Oct 2024); uses a small diesel pilot to ignite a premixed H2/air charge; Yanmar claims the world's highest hydrogen fuel ratio for this class (ratio value not stated in the release located). Hydrogen-engine production plan approved 9 Jan 2025 under Japan's zero-emission-ships programme.	https://www.yanmar.com/global/news/2024/10/30/143738.html https://www.yanmar.com/global/about/ycg/news/2025/01/27/147069.html
DEUTZ	No known programme.	No known programme.	Series H2 genset engine in the field. TCG 7.8 H2 hydrogen engine, 200 kW output, derived from the DEUTZ diesel platform; genset pilot with RheinEnergie delivered up to 170 kVA electrical output; volume production stated to start 2024 ; ~100 hydrogen-powered generator sets commissioned in China (first units in operation). Sub-MW class -- relevant for auxiliary/backup rather than utility duty.	https://www.deutz.com/en/products/hydrogen-engines/ https://www.deutz.com/en/news/press-releases/news-detail/deutz-puts-hydrogen-powered-generator-sets-into-operation-in-china/
Capstone (Capstone Energy+ / Capstone Green Energy)	Microturbine class -- 100 % H2 run . Whole product line made 30 % H2-blend compatible (announced 29 Mar 2022). C65 microturbine run on 100 % hydrogen using Capstone's patented High Flame Speed injector , with high combustion stability and injector integrity reported; up to 70 % H2 blend ran in an off-the-shelf microturbine before any hardware/software change. 100 %-H2 microturbine pilot reported in Australia. Unit sizes 65 kW-1 MW class -- distributed/auxiliary scale, not utility scale .	No known programme.	No known programme.	https://ir.capstonegreenenergy.com/press-releases/detail/3799 https://www.businesswire.com/news/home/20220329005359/en/Capstone-Green-Energy-Achieves-Goal-of-30-Hydrogen-Blend-Compatible-Products

Cross-cutting read (derived from the table only)

• **Gas turbines are furthest ahead on *large* hydrogen.** Every major frame OEM (GE Vernova, Siemens Energy, MHI, Ansaldo, Doosan) has a **commercial blend rating today **(typically 30-75 vol %)** **and a** 100 % roadmap landing

~2028-2030. **The** only 100 %-H2 machines that are actually released or near-released are small/industrial**: Kawasaki GPB17MMX (1.8 MW, dry), Baker Hughes NovaLT16 (~16 MW), GE Vernova B/E-class DLN retrofit (from 2026), GE Vernova LM6000VELOX (Whyalla, commissioning early 2026).

• **The 4-stroke engine side has genuinely caught up at MW scale.** Wärtsilä's **31H2** running on **100 % H2 into the Spanish grid (Jun 2026)** at **4.6-10.4 MW/unit** is the single most decision-relevant 2026 datapoint for a modular hydrogen power block -- it has no ammonia equivalent (see repo `NH3\_gas\_turbine\_vs\_gas\_engine\_comprehensive\_comparison.md`, where the ammonia engine is still marine-first).

• **The 2-stroke column is nearly empty, and is marine-only where it exists.**

Only **J-ENG (6UEC35LSGH, >95 % H2 energy at full load, Mar 2026)** and the **MAN B&W ME-GI single-cylinder conversion by MITSUI (2024)** exist, both as demonstrations aimed at **shipping in the 2030s** -- **no stationary power-generation 2-stroke hydrogen product exists from any OEM.**

• **Distinguish dry (DLN/DLE) from diffusion firing.** Several "100 % hydrogen" claims are **diffusion-flame** results (e.g. Kawasaki Kobe 2020), which normally carry a NOx/diluent penalty; the commercially meaningful milestones are the **dry** ones (Kawasaki GPB17MMX 2023; GE Vernova B/E-class DLN <25 ppm NOx, Jan 2025). Do not merge the two categories.

• **Retrofit is a distinct competitive lane.** Hanwha/PSM **FlameSheet** and GE Vernova's B/E-class DLN are aimed at the **installed fleet**, not new units -- relevant if any MYSGH2 permutation contemplates converting an existing Singapore/Johor gas turbine rather than buying new.

Methodology

• **Search-verified 2026-09-10** via web search across OEM press rooms (GE Vernova, Siemens Energy, MHI/Mitsubishi Power, Ansaldo Energia, Baker Hughes, Rolls-Royce, Solar Turbines, Caterpillar, Everllence/MAN ES, WinGD, Wärtsilä, Kawasaki, J-ENG, HD Hyundai, Doosan, INNIO, Cummins, Hanwha/PSM, Bergen, Yanmar, DEUTZ, Capstone) and trade press (POWER Magazine, Turbomachinery Magazine, Gas Turbine World, Power Engineering International, Hydrogen Insight, Hydrogen Central, The Motorship, Riviera Maritime, Power Progress, Offshore Energy, Modern Power Systems).

• **Prior repo sourcing reused, not re-derived:** the GE 9HA.02 H2 roadmap and H2-CCGT capex basis (memory.md §3/§6), the MHI JAC 30 %->50 % H2 milestones and H-25 frame data (memory.md §6 and `power\_generation/NH3\_gas\_turbine\_vs\_gas\_engine\_comprehensive\_comparison.md` §2

rows M6/M3), and the Baker Hughes/Hanwha NovaLT16 thread.

• **Limitation -- fetch blocking.** Several OEM domains

(`gevernova.com`, `siemens-energy.com`, `wartsila.com`, `innio.com`,
`powermag.com`, `gasturbineworld.com`, `fuelcellsworks.com`, `splash247.com`)

were **blocked by the network egress proxy** for direct page retrieval in this session. Their content was verified through **search-engine result summaries of those same pages, **and the canonical URLs are cited**. Any figure that a reader intends to use in an investment-grade deliverable should be re-opened at the source URL and confirmed.** This is a sourcing-strength caveat, not a fabrication: no number in the table originates from model memory alone.

• **"No known program"** in this table means *no programme located in this survey*. It is a statement about the search, not proof of absence.

Open questions / needs verification

• [] **Solar Turbines:** is there a released **100 % H2** product (vs. the sourced <=80 % field experience and 60 % Chevron blend)? Not located.

• [] **Hanwha:** the "***100 % hydrogen combustion in an 80 MW-class gas turbine**" claim on psm.com -- date, frame, dry vs. diffusion, NOx. Vendor claim only.

• [] **MHI H-25 100 % H2:** commercialisation status after the 2024 Takasago test -- no post-2024 update located (mirrors the same silence flagged for the H-25 **ammonia** programme in the repo's NH3 comparison, §2 row M5).

• [] **HD Hyundai HiMSEN pure-hydrogen 4-stroke:** was the Dec-2025 sales release actually made? Trade-press plan only.

• [] **Cummins:** any hydrogen **genset / stationary power-gen** offering, as distinct from on-/off-highway ICE?

• [] **Everllence/MAN turbomachinery (MGT series):** hydrogen capability not surveyed.

• [] **Toshiba Energy Systems, IHI, MTT, OPRA, Ethos:** not surveyed for hydrogen gas turbines this pass (IHI's located programme is **ammonia**, not hydrogen).

• [] **Basis discipline:** confirm **vol % vs. energy %** for each engine-side figure (Kawasaki 30 %, INNIO 25 %, Bergen 25 %, MAN 25 %) before any cross-OEM ranking -- the table reproduces each source's own wording.

• [] **Efficiency and NOx on hydrogen** are *not* in this table (the survey is a who's-who of availability). Net LHV efficiency, heat rate and NOx/ppm on H2 per machine remain to be collected before this can feed the P1 (hydrogen-to-Singapore) power-block comparison.

• [] **Ambient derate at tropical (Johor/Singapore) conditions** on hydrogen -- not published by any OEM located.

Sources (canonical URLs cited in the table)

Gas turbines

- GE Vernova -- 100 % hydrogen DLN combustor validation, B-/E-class (15 Jan 2025): <https://www.gevernova.com/news/press-releases/ge-vernova-validates-100-hydrogen-fueled-dln-combustor-technology-aiming-6>
- Turbomachinery Magazine -- GE Vernova DLN combustor passes validation with 100 % hydrogen: <https://www.turbomachinerymag.com/view/ge-vernova-s-dln-combustor-passes-validation-test-with-100-hydrogen>
- POWER Magazine -- GE Vernova unveils 100 % hydrogen-fuelled aeroderivative solution (LM6000VELOX, ATCO/BOC Whyalla): <https://www.powermag.com/ge-vernova-unveils-100-hydrogen-fueled-aeroderivative-gas-turbine-solution-secures-first-customer/>
- GE Vernova -- 7HA / 9HA product pages (50 % H2 today, pathway to 100 %): <https://www.gevernova.com/gas-power/products/gas-turbines/7ha> ; <https://www.gevernova.com/gas-power/products/gas-turbines/9ha>
- Siemens Energy -- HYFLEXPOWER: gas turbine (SGT-400) operated on 100 % renewable hydrogen, world first: <https://www.siemens-energy.com/global/en/home/press-releases/hyflexpower-consortium-successfully-operates-a-gas-turbine-with-.html>
- HYFLEXPOWER project -- first successful demonstration with 100 % green H2 (12 Oct 2023): <https://www.hyflexpower.eu/2023/10/24/first-successful-demonstration-with-100-green-h2/>
- Siemens Energy -- SGT-800 (up to 75 vol % H2; TÜV SÜD "H2-Ready" plant concept): <https://www.siemens-energy.com/global/en/home/products-services/product/sgt-800.html>
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- Mitsubishi Heavy Industries -- 30 % hydrogen co-firing, grid-connected JAC at T-Point 2 (Nov 2023): <https://www.mhi.com/news/23113001.html>
- Gas Turbine World -- Mitsubishi Power hydrogen gas turbine technology (30 %->50 %->100 % by 2030; H-25 100 % H2 test 2024): <https://gasturbineworld.com/mitsubishi-power-hydrogen-gas-turbine-technology/>
- Mitsubishi Power Americas -- Takasago Hydrogen Park: <https://power.mhi.com/regions/amer/success-stories/hydrogen-park>
- Ansaldo Energia -- GT36 sequential combustion achieves 0-100 % hydrogen (FLEX4H2): <https://www.ansaldoenergia.com/about-us/media-center/power-generation-news-insights/detail-news/gt36-sequential-combustion-technology-achieves-100-hydrogen>
- Fuel Cells Works (1 Sep 2026) -- Ansaldo GT36 gains TÜV SÜD hydrogen-readiness certification; AE94.3A HVO validation: <https://fuelcellsworks.com/2026/09/01/h2/ansaldo-energia-gt36-gains-hydrogen-readiness-certification-as-ae94-3a-passes-hvo-validation>
- Ansaldo Energia -- AE94.3A (co-firing up to 40 % H2; 25 % H2 units in service since 2006, >300,000 EOH): <https://www.ansaldoenergia.com/offering/equipment/turbomachinery/heavy-duty-gas-turbine/ae943a>
- Baker Hughes -- NovalT16 (start-up and operation to 100 % H2, no hardware change, no pilot fuel): <https://www.bakerhughes.com/gas-turbines/novalt-technology/novalt16>
- Gas Turbine World -- Baker Hughes readies NovalT turbine for 100 % hydrogen (Florence test facility; Air Products Edmonton): <https://gasturbineworld.com/baker-hughes-hydrogen-turbines/>
- Solar Turbines -- Converting high-hydrogen fuel to electricity (since 1985, >2 M hours, up to 80 % H2 content): https://www.solarturbines.com/en_US/about-us/news-and-press-releases/converting-high-hydrogen-fuel-to-electricity.html
- Turbomachinery Magazine -- Vendor spotlight: Solar Turbines (Taurus 60 / Titan 130 high-H2 fleet; 60 % blend with Chevron, 2024): <https://www.turbomachinerymag.com/view/vendor-spotlight-solar-turbines>
- Kawasaki -- Hydrogen co-firing gas engine / hydrogen product portfolio: https://global.kawasaki.com/en/energy/equipment/gas_engines/hydrogen_blended.html
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- Gas Turbine World -- Doosan Enerbility's new 380 MW H-class turbine (DGT6-300H.S1/S2; KIMM >=50 % H2 combustor; 400 MW 100 % H2 target 2027): <https://gasturbineworld.com/doosan-gas-turbines/>
- POWER Magazine -- South Korea eyes first hydrogen gas turbine demonstration in 2027: <https://www.powermag.com/south-korea-eyes-first-hydrogen-gas-turbine-demonstration-in-2027/>
- Hanwha -- world-first 59.5 % hydrogen co-firing in an 80 MW-class gas turbine (Daesan): <https://www.hanwha.com/newsroom/news/press-releases/hanwha-becomes-worlds-first-to-successfully-demonstrate-a-59-5-hydrogen-co-firing-rate-in-80-megawatt-class-gas-turbines.do>
- PSM (Hanwha Power) -- PSM, Thomassen Energy and Hanwha Power Systems achieve record hydrogen use (FlameSheet, 60 % H2 baseload, single-digit ppm NOx/CO): <https://www.psm.com/news-press-release-archive/psm-thomassen-energy-and-hanwha-power-systems-achieve-record-hydrogen-use-in-gas-turbine-power-plant> ; <https://www.psm.com/clean-energy>
- Capstone -- significant progress in microturbine hydrogen testing (C65 on 100 % H2, High Flame Speed injector): <https://ir.capstonegreenenergy.com/press-releases/detail/3799> ; 30 % H2-blend-compatible product line (Mar 2022): <https://www.businesswire.com/news/home/20220329005359/en/Capstone-Green-Energy-Achieves-Goal-of-30-Hydrogen-Blend-Compatible-Products>

Reciprocating engines -- 4-stroke

- Wärtsilä -- world's first large-scale 100 % hydrogen engine tested at Bermeo, supplying the Spanish grid (11 Jun 2026): <https://www.wartsila.com/media/news/11-06-2026-world-s-first-large-scale-100-hydrogen-engine-tested-at-wartsila-s-bermeo-laboratory-to-support-the-spanish-grid-3760292>
- Wärtsilä -- launch of the world's first large-scale 100 % hydrogen-ready engine power plant (31H2 / 31SG-H2; tendering 2025, delivery 2026), 18 Jun 2024: <https://www.wartsila.com/media/news/18-06-2024-wartsila-launches-world-s-first-large-scale-100-hydrogen-ready-engine-power-plant-3464281>
- Wärtsilä -- hydrogen-ready power plant solutions: <https://www.wartsila.com/energy/sustainable-fuels/hydrogen-power-plant> ; Wärtsilä 31 platform data (bore/stroke, 8-16 cyl, 4.6-10.4 MW): <https://www.wartsila.com/marine/products/engines-and-generating-sets/wartsila-31>
- INNIO -- Jenbacher gas engines ready for hydrogen (Type 4 on 100 % H2; "Ready for H2" 25 vol %; full line from 2025): <https://www.innio.com/en/news-media/press-releases/innio-jenbacher-gas-engines-ready-for-hydrogen/>
- INNIO + Net Zero Innovation Hub -- world-first 3 MW 100 %-hydrogen data-centre backup demonstration (23 Apr 2026): <https://www.innio.com/en/news-media/press-releases/innio-and-net-zero-innovation-hub-deliver-world-first-3-mw-demonstration-of-100-hydrogen-backup-power-for-data-centers/>
- INNIO -- technology selected for first 100 % hydrogen engine power plant in Asia Pacific (Hyosung, Ulsan): <https://www.innio.com/en/news-media/press-releases/innio-technology-selected-for-first-100-hydrogen-engine-power-plant-in-asia-pacific/>
- Rolls-Royce -- H2-ready certificate (TÜV SÜD) for mtu Series 4000 FNER/FV gas engines, 25 vol % blend and 100 vol % operation (11 Jul 2024): <https://www.rolls-royce.com/media/press-releases/2024/11-7-2024-energy-transition-rr-reaches-milestone-with-h2-ready-certificate-for-mtu-engines.aspx>
- Rolls-Royce -- fast-start mtu gas gensets for data centres (Series 4000 L64, 2.8 MW, 45 s, from 2026): <https://www.rolls-royce.com/media/press-releases/2025/02-10-2025-rr-introduces-fast-start-mtu-gas-gensets-for-powering-data-centers.aspx>
- Rolls-Royce -- mtu engines successfully tested with pure hydrogen (Jan 2023) / Enerport II 100 % H2 CHP: <https://www.rolls-royce.com/media/press-releases/2023/16-01-2023-rr-successfully-tests-mtu-engines-with-pure-hydrogen.aspx>

- Caterpillar -- Cat CG260 gas generator set added to hydrogen lineup: https://www.cat.com/en_US/news/engine-press-releases/caterpillar-advances-hydrogen-program-by-adding-cat-cg260-gas-generator-set-to-lineup.html
- Caterpillar -- hydrogen-fuelled power solutions, gensets and retrofit kits 600 kW-2.5 MW; G3516 100 % H2 demonstrator (1,000 kW): https://www.cat.com/en_US/news/engine-press-releases/caterpillar-expands-lineup-of-hydrogen-fueled-power-solutions-with-generator-sets-and-upgrade-kits.html
- Everllence / MAN Energy Solutions -- hydrogen power solutions (35/44DF H2 research engine; 25 % H2 blend on 35/44GTS and 51/60G): <https://www.man-es.com/energy-storage/campaigns/hydrogen-power-solutions>
- Kawasaki -- world's first commercial launch of a hydrogen 30 % co-firing large gas engine system (30 Sep 2025; 8 MW-class verification at Kobe Works): https://global.kawasaki.com/en/corp/newsroom/news/detail/?f=20250930_4865
- Bergen Engines -- 100 % hydrogen-fuelled engine on track (Dec 2024): <https://www.bergenengines.com/bergen-engines-100-hydrogen-fueled-engine-on-track/> ; increase in blended hydrogen offering (25 % without modification): <https://www.bergenengines.com/increase-in-blended-hydrogen-offering/>
- Modern Power Systems -- Bergen demonstrates 100 % hydrogen: <https://www.modernpowersystems.com/analysis/bergen-demonstrates-100-hydrogen/>
- Yanmar -- successful operation at rated output (~500 kW) of a hydrogen 4-stroke high-speed engine for coastal vessels (30 Oct 2024): <https://www.yanmar.com/global/news/2024/10/30/143738.html>
- Yanmar -- hydrogen engine production plan approved under the zero-emission ships project (9 Jan 2025): <https://www.yanmar.com/global/about/ygc/news/2025/01/27/147069.html>
- DEUTZ -- hydrogen engines (TCG 7.8 H2, 200 kW): <https://www.deutz.com/en/products/hydrogen-engines/> ; hydrogen-powered generator sets in operation in China: <https://www.deutz.com/en/news/press-releases/news-detail/deutz-puts-hydrogen-powered-generator-sets-into-operation-in-china/>
- The Motorship -- HiMSEN confirms 2025 pure-hydrogen engine launch plans: <https://www.motorship.com/2-stroke-and-4-stroke/himsen-confirms-2025-pure-hydrogen-engine-launch-plans/1488666.article>
- HD Hyundai -- HD Hyundai Infracore hydrogen engine business roadmap (power generation 2026; 11 L H2 engine): <https://esg.hd.com/en/news/842>
- Cummins -- B6.7H hydrogen engine development completed (Route One): <https://www.route-one.net/suppliers/cummins-b6-7h-hydrogen-engine-development-is-completed/>

Reciprocating engines -- 2-stroke

- Offshore Energy -- MAN ES and MITSUI E&S perform world's first hydrogen combustion test on a marine 2-stroke engine (ME-GI, single cylinder, to 100 % load, up to 95 % GHG cut): <https://www.offshore-energy.biz/man-es-and-mitsui-es-perform-worlds-first-hydrogen-combustion-test-on-marine-2-stroke-engine/>
- Ocean News & Technology -- MITSUI performs world-first hydrogen test with MAN B&W two-stroke engine: <https://oceannews.com/news/science-technology/mitsui-performs-world-first-hydrogen-test-with-man-b-w-two-stroke-engine/>
- J-ENG -- world's first land-based operation of a marine hydrogen (two-stroke) engine, press release 28 Oct 2025 (6UEC35LSGH; HyENG JV with Kawasaki and Yanmar Power Technology): https://www.j-eng.co.jp/en/news/lnhnoq00000005xn-att/J-ENG_Press_20251028_EN
- Interesting Engineering -- world-first hydrogen-fuelled engine for large ocean ships (all-cylinder co-firing from Mar 2026; H2 >95 % of fuel energy at full load): <https://interestingengineering.com/energy/world-first-hydrogen-fueled-engine-cargo-ships>
- WinGD -- first ethanol-fuelled two-stroke in 2026 (context: current alternative-fuel roadmap = LNG/methanol/ethanol/ammonia): <https://wingd.com/news-media/news/wingd-to-introduce-first-ethanol-fuelled-marine-two-stroke-engine-in-2026>
- Power Progress -- WinGD looks to future fuels for ocean-going ships (hydrogen characterised as a building block rather than a direct marine fuel): <https://www.powerprogress.com/news/wingd-looks-to-future-fuels-for-ocean-going-ships/8024800.article>

Survey compiled 2026-09-10. Every quantitative cell is attributed to a linked source or explicitly marked unconfirmed / not located (CLAUDE.md §7). No efficiency, heat-rate, NOx-in-service, CAPEX or derate figure has been assigned to any machine in this document -- those were not sourced in this pass and are listed as open questions rather than estimated.